

FPGA Based Implementation of BPSK and QPSK Modulators using Address Reverse Accumulators

Amean Al safi and Bradley Bazuin
Electrical and Computer Engineering
Western Michigan University
Kalamazoo, Michigan, USA

Emails: {ameansharea.ghazi@wmich.edu ;brad.bazuin @wmich.edu}

Abstract— Implementation of digital modulators on Field Programmable Gate Array (FPGA) is a research area that has received great attention recently. Most of the research has focused on the implementation of simple digital modulators on FPGAs such as Amplitude Shift Keying (ASK), Frequency Shift Keying (FSK), and Phase Shift Keying (PSK). This paper presented a novel method of implementing Quadrature Phase Shift Keying (QPSK) along with Binary PSK (BPSK) using accumulators with a reverse addressing technique. The implementation of the BPSK modulator required two sinusoidal signals with 180-degree phase shift. The first signal was obtained using Look Up Table(LUT) based on Direct Digital Synthesizer (DDS) technique. The second signal was obtained by using the same LUT but after inverting the most significant bit of the accumulator to get the out of phase signal. For the QPSK modulator, four sinusoidal waves were needed. Using only one LUT, these waves were obtained. The first two wave were generated by using two accumulators working on the rising edge and the falling edge of a perfect twice frequency square wave clock which results in a 90-degree phase shift. The other waves were obtained from the same accumulators after reversing the most significant bit in each one. The implementation of the entire systems was done in the Very high speed integrated circuit Hardware Description Language (VHDL) without the help of Xilinx System Generator or DSP Builder tools as many papers did.

Keywords: *PSK; BPSK; QPSK; FPGA; VHDL; SDR; Digital Modulators*

I. INTRODUCTION

The concept of Software Defined Radio (SDR), which was first presented by Joseph Mitola [1]- [3] as a way of implementing programmable and reconfigurable radio transceivers, has become a widely used method in implementing various communication systems. The advantages of SDR-based systems have become more apparent with the tremendous development in low cost hardware terminals. These reconfigurable terminals hardware such as Universal Software Radio Peripheral (USRP) are the main components in any SDR-based system. The USRP itself consists of a FPGA, Digital-to-Analog Converter (DAC),

Analog-to-Digital Converter (ADC), and mixers, with the FPGA performing critical filtering and data rate processing work for SDR-based system. Extending this concept to incorporate additional modulation and demodulation operations, is it possible to build SDR-based systems using FPGAs, a simple or even embedded CPU, and other low cost electronics? The answer to this questions can be found be looking at some of the research activity in this direction that focuses on the implementation of digital modulators using FPGA.

Implementation of digital modulators on FPGAs has experienced numerous research contributions in the last few years but there is still significant work that needs to be done. Most of the previous work has focused on the implementation of simple digital modulators such as ASK, FSK, and PSK. Quadri and Tete presented a review of the main research papers in this area with some implementation examples using Xilinx System Generator [4]. Some of this research will be summarized here, especially those related to the work being reported.

In 2014, Bhore and Sarde presented an implementation of a BPSK modulator and demodulator on a SPARTAN -3 FPGA. They used MATLAB SIMULINK to build their entire system and Xilinx System Generator to generate the VHDL implementation of the models [5]. BASK, BFSK, and BPSK modulators were implemented on an Altera development and education board using Verilog by Erdogan et al [6]. They used DDS to generate the sine wave carrier at a frequency of 1KHz. Their system was implemented directly in Verilog without using Xilinx System Generator tools. BPSK was also implemented on SPARTAN -3 starter kit board by Popescu et al [7]. MATLAB SIMUINK and Xilinx System Generator were used as tools to generate the VHDL implementation module. Chye et al presented a detailed guideline on how to design and implement BPSK transmitter on Virtex-4 FPGA development board with DAC available on a P240 analogue module. Software and design tools such as MATLAB SIMULINK and Xilinx System Generator were used to verify the design output in terms of behaviour, functionality, synthesis, timing, and constraints area. The implementation was done in Verilog which was generated by Xilinx System Generator [8].

Rajaram and Gayathre presented an FPGA based implementation of different digital communication schemes such as Quadrature Amplitude Modulation (QAM), BASK, BFSK, and BPSK. A SPARTAN-3 FPGA was used as the base of the implementation [9]. A COordinate Rotation DIgital Computer (CORDIC) algorithm was used to generate the sine wave carrier. The unique thing about this paper is that everything was done in VHDL not in MATLAB and Xilinx System Generator as in other papers. Zhuan Ye et al extended the concept of software defined functionalities to radio frequency with a novel "all-digital transmitter"[10]. The all-digital transmitter uses pulse width modulation (PWM) method such that the Radio Frequency (RF) signal with binary format can be directly synthesized in the digital domain. The low rate of the signal processing of PWM makes it suitable for implementation using FPGA. By combining a universal QAM modulator and an RF pulse width modulator, they demonstrated a real time QAM transmitter system with digital RF output using a realistic signalling format.

Not only digital modulators, as it was explained in the last few paragraphs, but also analog modulators have been implemented using a variety of FPGA based development boards [11]- [17]. Some of these researches have reached what they consider optimum solutions in term of efficiency, power consumption, and resources utilization.

Despite all the progress that has been made, there is still work needs to be done. This work will focus on implementing more complex modulation schemes, looking for more efficient designs for the current implemented ones, and developing solutions not requiring proprietary software tools or libraries. In this paper, we presented a novel method of implementing BPSK and QPSK modules on FPGA using accumulators with a reverse addressing technique and one LUT. To date, no one has presented or used this idea before in term of FPGA based modulators. The rest of this paper is organized as follow: section II presents a review of the research work in this direction, section III illustrates the proposed implementation methods, section IV is the implementation results, and finally section V is the conclusion and future work.

II. RELATED WORK

The successful implementation of simple communication modulators has opened the doors for the implementation of more complicated systems such as QPSK and QAM. This paper focused on the implementation of BPSK and QPSK using a reverse addressing accumulators. Several papers existed in term of implementing BPSK using FPGA but none of them has mentioned the technique of using reverse address accumulator to get out of phase sinusoidal signals. In addition, there are very few research papers that have investigated the implementation of QPSK modulators using FPGA.

In 2015, Gaikwad et al presented an implementation of QPSK transmitter and receiver system on a FPGA. The Concept they presented in their paper is similar but uses a different approach [18]. Song and Yao also presented an FPGA based QPSK modem using a FPGA. They do not show or report on generating an analog or RF signal output, but directly connect digital I and Q modem outputs to I and Q received inputs. Overall, not enough detail was given in term

of utilizations and implementation [19]. Thombare and Shah presented an efficient way of implementing QPSK modulator using 16 and 64 ROM memory [20]. They depend on the current and next values of I (in-phase component) and Q (quadrature component) data to produce the next symbol from the symbols stored in a ROM table. Even though they did not mention it, their concept is very similar to the previously patented Feher QPSK modulation technique [21]. Kazaz et al presented an efficient design and implementation of SDR based QPSK modulator on an Altera Cyclone IV FPGA [22]. For the design and implementation of a QPSK modulator they used Altera DSP Builder Tool combined with MATLAB Simulink, Modelsim and Quartus II design tools. As reconfigurable hardware platform they used an Altera DE2-115 development and education board with ADC/DAC daughter card. They created their project in MATLAB SIMULINK, then they used Altera DSP Builder Tool to convert the system to VHDL. Elamary et al presented a VHDL based implementation of QPSK modulator on Altera board. They used phase shifters to generate four signals from one input sine wave [23]. Those signal were used as inputs to a multiplexer which select one of them based on the message signal. They compare their system with a simulated model in MATLAB before the practical test. The generated QPSK signal was cleaned by an analog filter before the transmission. Kolankar and Sakhare presented an efficient implementation in term of power consumption of QPSK modulator using Xilinx System Generator [24]. They designed the whole system in MATLAB SIMULINK, then they used Xilinx System Generator to generate the VHDL implementation of the model. They used hardware co-simulation which is a new feature in MATLAB by the virtue of Xilinx System Generator to make sure the implemented system is efficient in term of performance and hardware implementation.

The way we implemented our systems is novel and different from what others presented as it will be shown in the next section. We used one LUT and one clock signal and we worked with the accumulator output to generate different signals at various phase shifts.

III. PROPOSED IMPLEMENTATION METHOD

A. BPSK:

BPSK is the simplest form of PSK. It uses two signals with 180-degree phase shift to generate its symbol. Fig.1 is the constellation diagram of BPSK. The general equation of BPSK is [25]:

$$S_n(t) = \sqrt{\frac{2E_s}{T_s}} \cos(2\pi f_c t + (1-n)\pi) \quad n = 0,1 \quad (1)$$

Where E_s is the energy per symbol, T_s is the symbol duration, and f_c is the carrier frequency. Based on the value of n , two signals can be generated:

$$S_0(t) = -\sqrt{\frac{2E_s}{T_s}} \cos(2\pi f_c t) \quad \text{when } n = 0 \quad (2)$$

$$S_1(t) = \sqrt{\frac{2E_s}{T_s}} \cos(2\pi f_c t) \quad \text{when } n = 1 \quad (3)$$

This means two sinusoidal waveforms have to be generated in order to be able to implement BPSK in VHDL. For this purpose, a DDS technique was used. In the DDS method, a 24-bit accumulator with LUT were used for the sine wave generation. The accumulator works on the rising edge of the clock. The 8-most significant bits of the accumulator were used as an address to select the corresponding amplitude of the sinusoidal signal from the LUT. Since the used address has 8-bit width, the LUT has to have 256 samples values which cover one cycle of the sinusoidal signal. These 256 sample values were generated in MATLAB and exported into VHDL as a text file. They also can be generated by using other software such as Microsoft Excel. The accumulator and LUT can give us only one signal based on the previous description. For BPSK, we need to find a way to get the other signal which is 180-degree out of phase as compared to the first signal. The easy way to do that is to multiply the first signal by (-1) which is corresponding to 180-degree phase shift. Unfortunately, in VHDL, programmers try to avoid multiplication as possible as they could due to high resources consumption. Another option is to invert or reverse the most significant bit in the accumulator. To explain how this technique was discovered, we assumed that the eight most significant bits of the accumulator start from “00000000” to “11111111” where “00000000” is the first address, “00000001” is the second address, and “11111111” is the 256th address. The first address is corresponding to zero-degree angle while last address will be corresponding to the angle $(\frac{360}{256} * 255)$ degree. The angle difference between any two adjacent addresses will be $\frac{360}{256}$. This means the angle 180 degree is corresponding to the address “10000000”. If we compare this address with address corresponding to the zero-degree angle (“00000000”), we discover the only difference they have is the most significant bit. Therefore, reversing the most significant bit of the accumulator gives a 180-degree out of phase signal as compared to the original signal. For flexibility and testing, this operation is performed by an exclusive or (XOR) logic gate function.

After getting the two out of phase sinusoidal signals, a multiplexer was used to generate the output BPSK signal as shown in Fig.2. The width of S0, S1, and the BPSK output was selected to be 16-bit. An 8-bit width can be used but we tried to get a higher precision output for driving a DAC. The block diagram of the BPSK implementation is shown in Fig.2.

B. QPSK

QPSK, which is also known as quadrature PSK, 4-PSK, or 4 QAM, uses four different points on the constellation diagram to represent its symbols. Each symbol can be encoded by using two bits. Fig.3 is the constellation diagram of QPSK. Mathematically speaking, QPSK can be used to double the data rate as compared to BPSK using the same bandwidth. This means that QPSK can transmit twice the data rate in a given bandwidth as compared to BPSK. Fig. 4 explains how to generate QPSK signal from a stream of data. The general form of QPSK symbol is [25]:

$$S_n(t) = \sqrt{\frac{2E_s}{T_s}} \cos(2\pi f_c t + (2n-1)\frac{\pi}{4}) \quad n=0,1,2,3 \quad (4)$$

Based on the value of n in equation 4, four different signals can be generated. These signals are 90-degree out of phase to each other. Hence, implementation of QPSK modulator required the generation of four sinusoidal signals that sequentially have a 90-degree phase shift from the previous signal. To do that, the first signal can be generated as it was described in the previous section using accumulator working on the rising edge of the clock and LUT. The second signal was generated using the same LUT but at this time another accumulator working on the falling edge of a 50% duty cycle clock was used. This accumulator generates a signal with 90-degree phase shift as compared to the first one. The other two signals are generated by reversing the most significant bit in the first and second accumulator and using the same LUT.

After the generation of the four signals, QPSK modulator can be implemented as a next step. The incoming binary data has to be converted from serial to parallel data as it is shown in Fig. 5. This process can be easily done in VHDL. The last step in the implementation is to use a multiplexer which is going to choose among the four generated sinusoidal signals based on the parallel version of the message signal as it is shown in the block diagram. S0, S1, S2, S3, and QPSK output signal were selected to have 16-bit width.

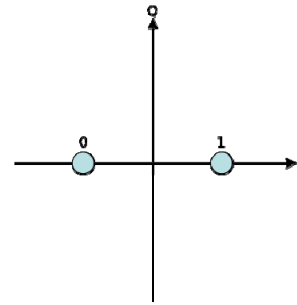


Fig. 1: BPSK constellation diagram.

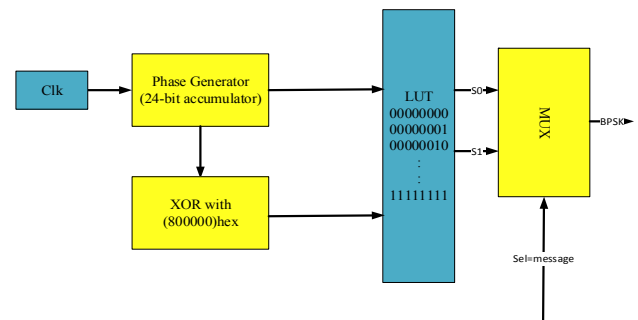


Fig. 2: Block diagram of the BPSK implementation in VHDL

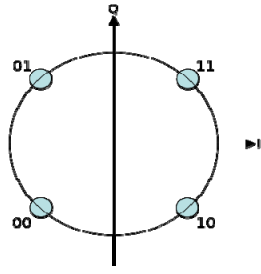


Fig.3: QPSK constellation diagram.

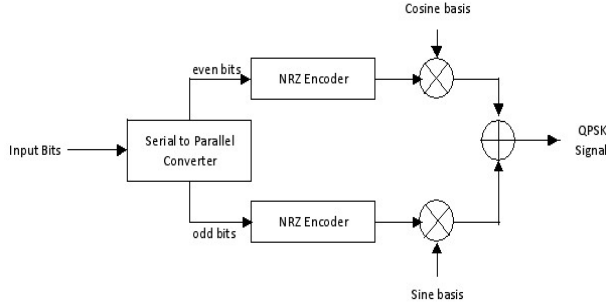


Fig.4: QPSK block diagram.

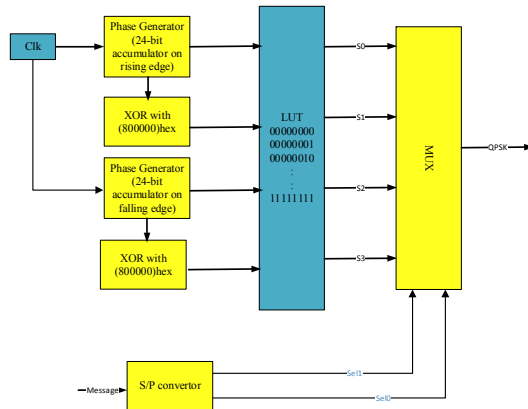


Fig. 5: Block diagram of the QPSK implementation in VHDL.

IV. IMPLEMENTATION RESULTS

After following the detailed descriptions of the implementation in VHDL, ModelSim and MATLAB were used as tools to visualize the output signals. For the BPSK part, the output waveform of the modulator was obtained using ModelSim based on the message signal ($m=010010011001000$). The implementation results were also exported into MATLAB as text files for both the generated waveforms and the BPSK output signal. The generated sinusoids are shown in Fig.6 which was plotted in MATLAB. It is very clear that the generated waves have 180-degree phase shift as compared to each other. The generated BPSK signal is shown in Fig.7. It is clear where the signal reversed its phase based on the incoming message.

For the QPSK part of the implementation, the detailed procedure in previous section was followed. Xilinx ISE was used as a software during the VHDL implementation while ModelSim and MATLAB were used as tools to visualize the output signals. The resultant QPSK signal was generated using

ModelSim based on the same input message in the BPSK part. The four generated sinusoidal waves were exported into MATLAB as text file to check if they meet the specifications we are looking for. It is clear that they met all the specifications in term of the 90-degree phase shift as shown in Fig. 8. The QPSK output signal was also exported as a text file to plot it in MATLAB which is shown in Fig. 9. We plotted the results in MATLAB as it is not easy to visualize the ModelSim results. To get a signal for transmission, a DAC and filtering process has to be included which we have left as a future work.

The primary differences between our implementation models and previous papers' models are the way we generated the signals at different phase shifts and the open-source tools we used for implementation purposes. We did not use phase shifters as in [23], or MATLAB SIMULINK and System Generator as in [19,24] or DSP Builder tools as in [22].

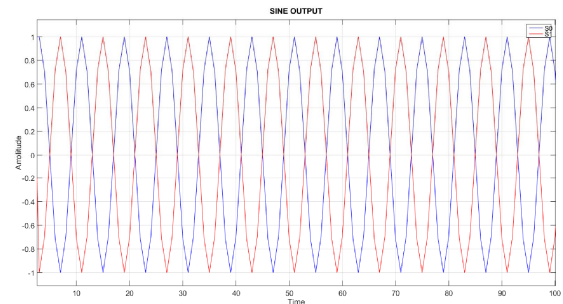


Fig. 6: The two generated out of phase sinusoids.

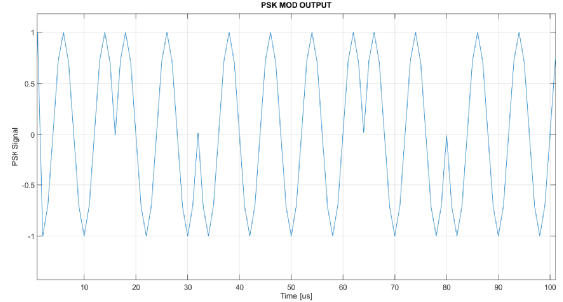


Fig. 7: BPSK implementation results.

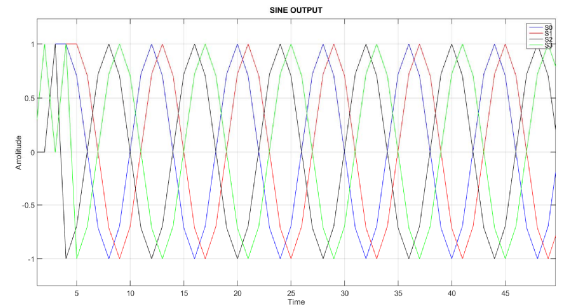


Fig. 8: The four generated sinusoids with 90-degree phase shift.

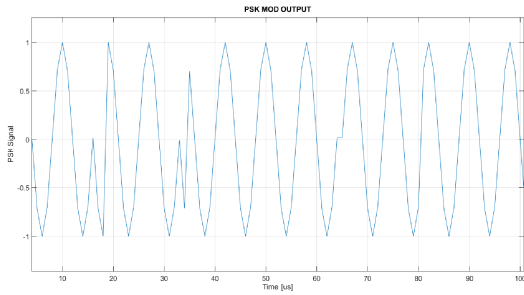


Fig. 9: QPSK implementation results.

V. CONCLUSION AND FUTURE WORK

An implementation of BPSK and QPSK modulation using accumulators with a reverse address technique was presented in this paper. For the implementation of the BPSK modulator, two sinusoidal signals with 180-degree phase shift were generated using a single LUT and without any multiplication instruction. The first signal was obtained using LUT and accumulator working on the rising edge of the clock based on DDS technique. The second signal was obtained by using the same LUT but after reversing the most significant bit of the accumulator to get a signal with 180-degree phase shift as compared to the first one. For the QPSK modulator, four sinusoidal waves were generated using only one LUT. The first two wave were obtained by using two accumulators working on the rising edge and the falling edge of the clock which makes them having 90-degree phase shift. The other waves were obtained from the first two accumulators after reversing the most significant bit. The implementation was done entirely in VHDL without the help of Xilinx System Generator or DSP Builder tools as in many papers.

As a future work, the authors will investigate the implementation of other digital modulators such as 8PSK, 16PSK, and higher order QAM. With successful implementations achieved, then a full range of modules can be available in a single FPGA or in-circuit FPGA programmability can be used in building signal processing blocks for SDR.

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